Welcome to CS/CNS/EE 156B (Learning Systems)!

- Policies (Yaser)
- The Project (Shrujana)
- Logistics (Shrujana)
- Techniques (Thursday)

Project Course

- Medical Application: Medical Image Diagnosis (CheXpert data set)
- Competing teams of 3 students each
- Three presentations/reports: 3rd, 6th, and 9th week
- Leaderboard to show progress of team submissions

What is NOT allowed

- No **CheXpert** data source is allowed other than what is provided by us on HPC. Don’t even look.
- No papers about the **CheXpert** data set or solutions are allowed.
- Any other clinical studies/data not related to **CheXpert** is fine.

What IS allowed

- Use any programming language you want.
- Use any technique or package you want that is not specific to CheXpert.
- Read any papers not involving CheXpert.
- Discuss freely, but only report your own work.

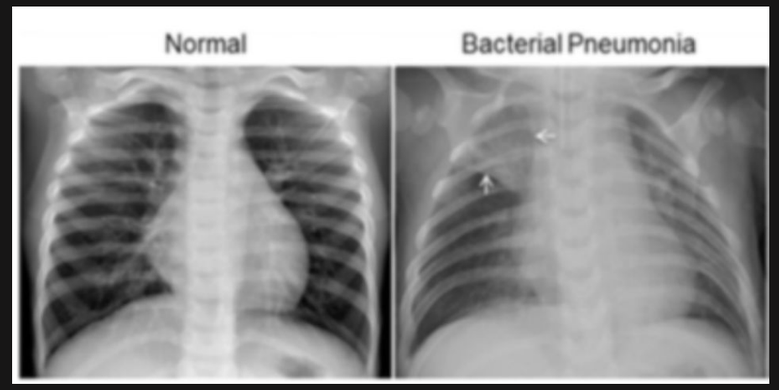
Grading Criteria

- The quality of + attendance at the 3 presentations
- Your ideas and techniques, even if they don’t work out
- The leaderboard score

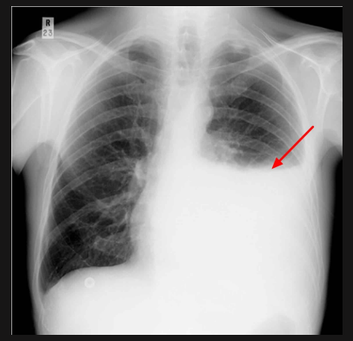
Chest X-ray Prediction!

Problem Setting

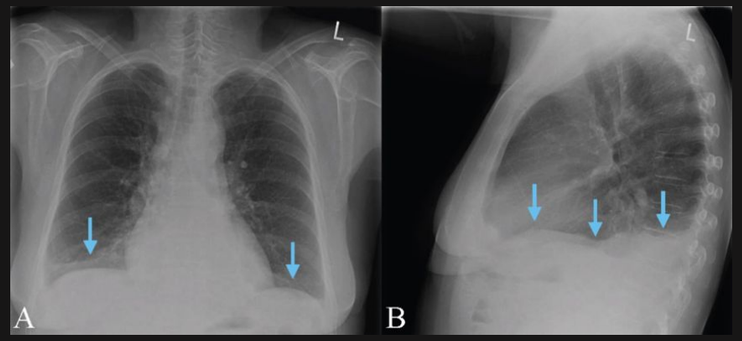
- Chest x-ray is the most common imaging examination globally
  - Screening, diagnosis, management of disease
- Radiologists "classify" x-rays
  - Training: ~13 yrs (4yr college + 4yr med. school + 4yr residency + 1yr specialty)
  - Cost: on average, > $150/hr ($300k/yr)



Pleural effusion



vs.



Dataset: CheXpert (Stanford, 2019)

224k chest X-rays across 65k patients

- Both frontal and lateral views

Labels

- Positive (1), uncertain (0), negative (-1), unmentioned (blank)
- Radiologist-annotated: very expensive => very few exist
- NLP auto-generated: cheap, noisier, but exist for all images
  - Extraction models trained on reports associated with X-rays
- Radiologist and auto-generated labels should correlate well

Prediction Task

Given an X-ray image, predict observation of ~~14~~ 9 pathologies

Trying to match auto-generated labels

- 80% dataset provided for training
- 10% dataset public holdout
- 10% dataset private holdout

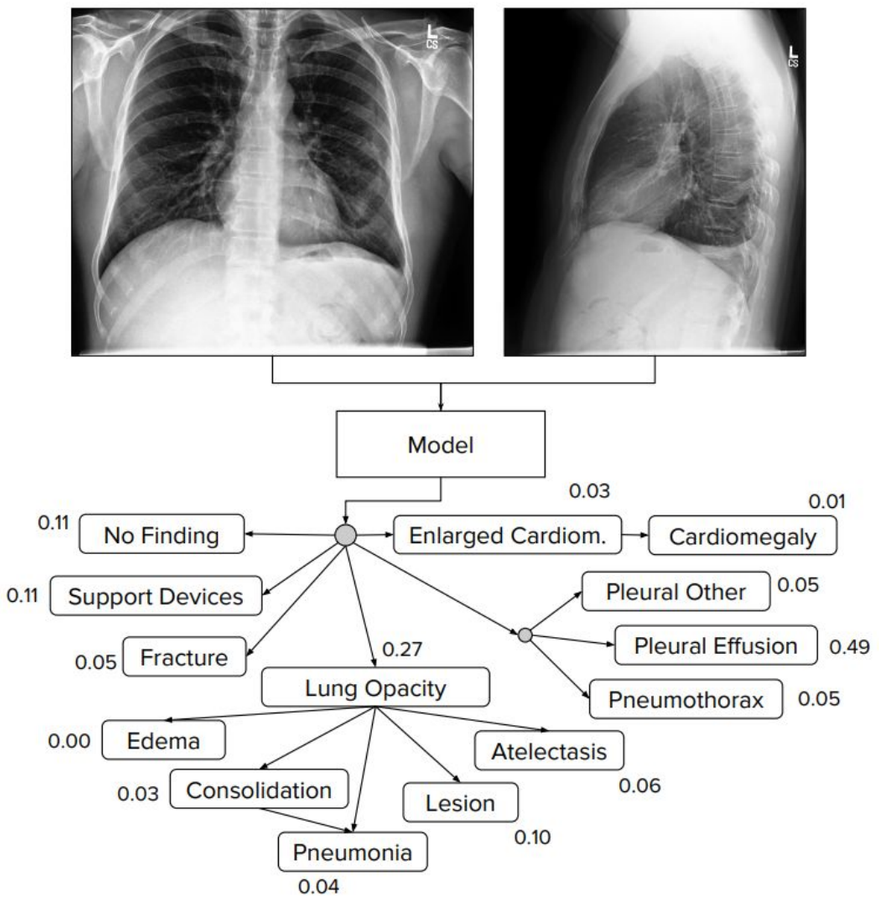


Figure 1: The CheXpert task is to predict the probability of different observations from multi-view chest radiographs.

Logistics

Leaderboard (Internal Site)

- Place for prediction submissions
- Teams of **3 members**
  - Can use Piazza to look for group members
  - Also, stay at end of class today if looking for group
  - **Fill out form w/ group members on Canvas/Piazza by this Friday (11:59 PM)**
- Public leaderboard
  - Bragging rights for the winners!
  - *Leaderboard has nonzero but small impact on grade!*
- **One submission per group is required by Sunday (11:59 PM)**
  - Just to make sure you are able to submit - can be any predictions as long as in the right format
  - One submission each week is *recommended* (but not required) to track your progress

Requirements

1. Each student should be present for their team's presentations (3rd, 6th, and 9th week).
2. Teams of 3 students each should be formed the first week, and each team should have a public leaderboard submission by the end of the first week **(April 5th, 11:59 PM)**
3. Each team should have a public submission by the 5th week **(May 1st, 11:59 pm PST)** where at least one pathology has normalized MSE score <= 0.99.

Live Presentations

- In person, Moore B270 (during this class time)
- Aim for 5-7 minutes total
- Assume your audience is familiar with the dataset and ML in general
- Focus on what you did in the last three weeks and how that's going (~80%) and what you plan to improve/try next (~20%)
- Attendance encouraged (even if not your own group!)

Individual Reports

- Each team member must submit ~1-page on presentation weeks
- Same requirements – focus on what you did, how that's going, and what you plan to improve/try next
- May include screenshots of code, sample results/visualizations, etc.
- Nothing to stress about!

Gradescope

- Submit PDF of slide deck after each of your presentations & individual reports
- Due on Friday at the end of the week you present
  - E.g., presentations on April 14th and 16th & reports will be due on Friday, April 17th at midnight

Course Staff & Office Hours

- Introductions on Thursday!
- Each TA will be available by request on Piazza or via email, and are happy to schedule on-demand in-person OH
- Will also be monitoring Piazza for any logistical questions

Computational Resources

- Caltech HPC
  - Tutorial / Setup document will be posted
  - More on this during Thursday lecture
  - You'll probably want to use GPU…

Questions?

Next time… (Thursday, 4/02)

- Idea primer–a few potential directions to look into from the TAs (~15-20 min)
  - Just to get you thinking!
  - Document will be published on Canvas/Piazza
- Submission process tutorial (~20 min)
  - Log in to submission site
    - Prepare a submission for the leaderboard (formatting)
  - Setup HPC
    - Run basic CNN example